

Eye

The eye is a photoreceptor organ.

It is found in a bony cavity in the skull which is called the orbit.

The wall of the eyeball is formed of 3 layers:

I) The outer fibrous layer:

It is a fibrous coat, formed of cornea anteriorly and sclera posteriorly. It is protective and supportive layer. It is formed of white fibrous CT.

II) The middle layer (uveal tract):

It is the vascular coat, formed of three parts

1-The iris.

2-The ciliary body.

3-The choroid.

III) The Inner layer (neural layer):

It is the nervous coat, formed of the retina.

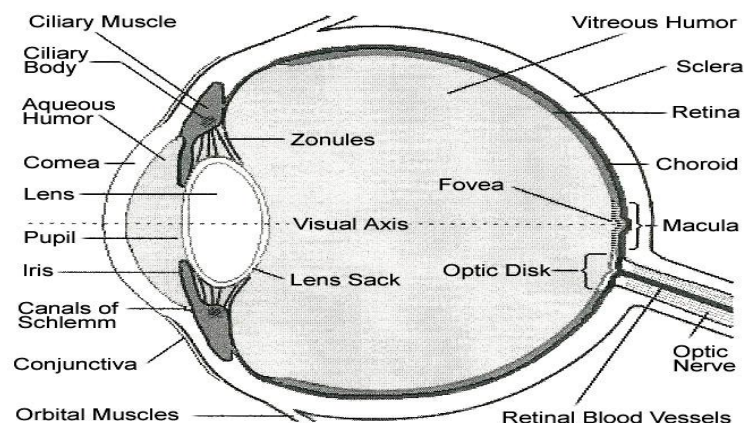
The eye contains three compartments:

1-The anterior chamber: between the cornea and the iris.

2-The posterior chamber: between the iris, ciliary body, and the lens.

3- The vitreous space: behind the lens.

- The anterior and the posterior chambers are connected by small opening in the iris called the pupil.
- Both chambers contain protein-poor fluid called aqueous humor, while the vitreous space is filled with a gelatinous material called vitreous body.



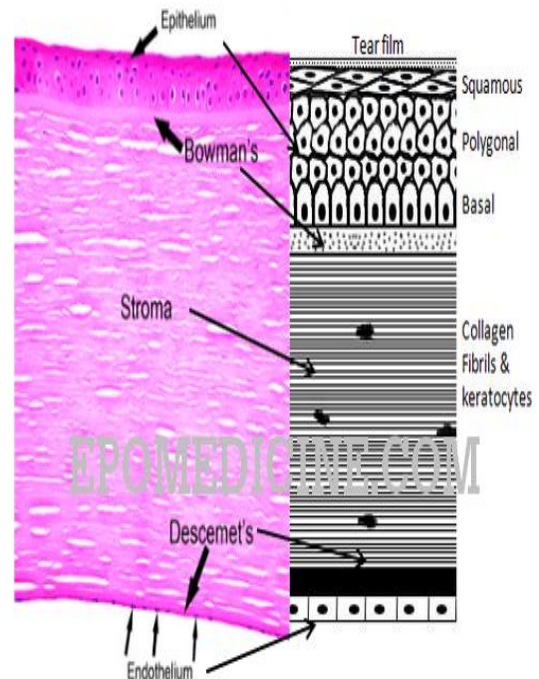
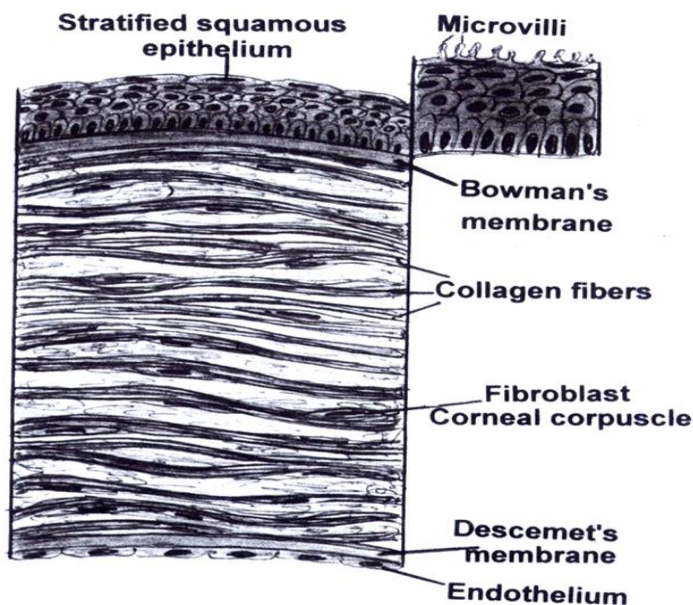
Outer fibrous layer:

A. Cornea: It forms the anterior 1/6 of fibrous layer. It is transparent “refracting media “that focuses image roughly on retina.

Structure of the cornea:

It is formed of three cellular layers separated by two acellular layers.

1. **Epithelium:** it is covered by stratified squamous non-keratinized epithelium. Rich in free nerve endings. Superficial cells contain microvilli embedded in tear film.
2. **Bowman's membrane:** it is **non-cellular** layer about 10–12µm thickness. It is formed of collagen fibers embedded in ground substance. **It acts as a protective layer against any trauma or bacterial invasion.**
So, its injury → scar formation → corneal opacity
3. **Stroma (substantia propria):** it is formed of many layers of parallel collagen bundles which are parallel to each other and run full width of the cornea. Between them, there are fibroblasts (keratocytes). The cells and the fibers are immersed in a substance rich in glycoproteins and chondroitin sulfate.
4. **Descemet's membrane:** it is **non cellular** (5- 10 µm) homogeneous layer. It is composed of fine collagenous filaments in three-dimensional network embedded in amorphous ground substance
5. **Corneal endothelium:** it lines the internal (posterior) surface of cornea. It is simple squamous endothelium rich in **mitochondria** and **microvesicles**. It is responsible for synthesis of proteins necessary for secreting and maintaining Descemet's membrane. The excess fluid within stroma resorbed by endothelium keeping stroma relatively dehydrated.



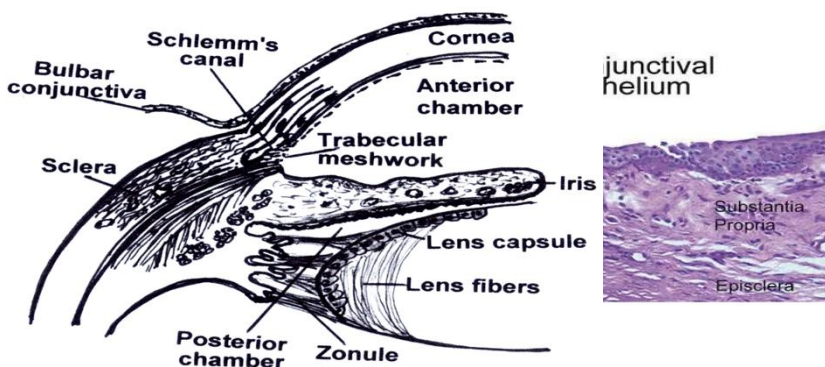
Why Cornea is transparent?

- 1- **Cornea is avascular structure (not penetrated by blood vessels).** It takes its nutrition from the aqueous humor and from the scleral capillaries in the cornea-scleral junction.
- 2- **Regular arrangement of collagen lamellae.**
- 3- **The same refractive index** of the collagen fibrils and the ground substance.
- 4- **Continuous withdrawal of water** from the cornea by Descemet's endothelial cells.

B. Sclera: It forms posterior 5/6 of outer fibrous layer. It is consisted of tough, dense CT with collagen bundles arranged in various directions, with moderate amount of ground substance, and few fibroblasts. It is opaque due to irregular arrangement of collagen fibers.

C. Limbus (cornea-scleral junction): It is highly vascular transition area between transparent cornea and opaque sclera.

1. Epithelial lining of cornea is continuous with that of bulbar conjunctiva.
2. Bowman's capsule: is replaced by sub-conjunctival loose CT which contains loop of blood vessels supply cornea with nutritive fluid.
3. Regular collagen fibers of corneal stroma are replaced by irregular fibers in sclera. At this point in stromal layer, irregular endothelium-lined channels "trabecular meshwork" merge to form Schlemm's canal, which drains fluid from anterior chamber of eye. Schlemm's canal communicates externally with venous system.
4. Descemet's membrane and endothelium of cornea disappear abruptly.

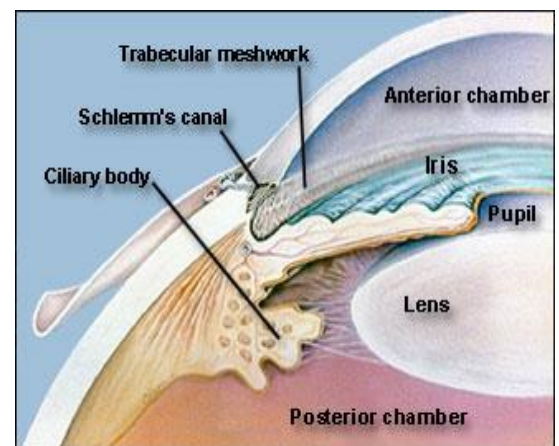
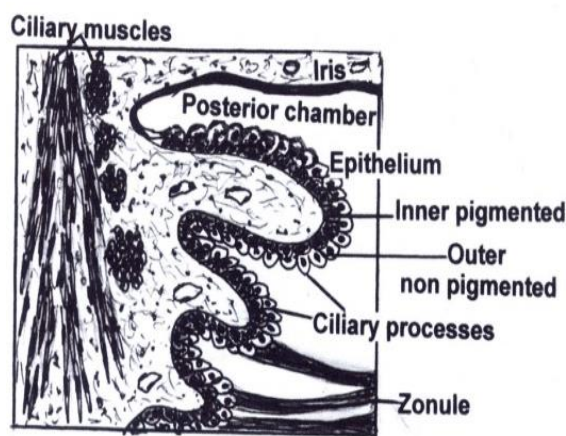


II. Middle or uveal layer of eye: Vascular layer of eyeball. It is formed of choroid, ciliary body and iris.

Ciliary body: Triangular in shape with its apex continuous posteriorly with choroid and its base faces the iris. Ciliary processes project from its medial border.

Function:

1. Secretion of the aqueous humor
 2. Give the attachment to the Suspensory ligament of the lens (Zonule)
- **Ciliary muscles:** Smooth muscle fibers arranged in two bundles radial and circular fibers. Contraction and relaxation of these muscles can change in the shape of the lens which is important in visual accommodation.

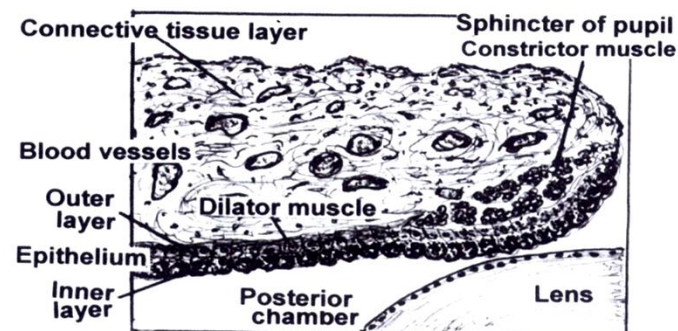


Iris:

The anterior part of the middle coat of the eye.

Function of iris:

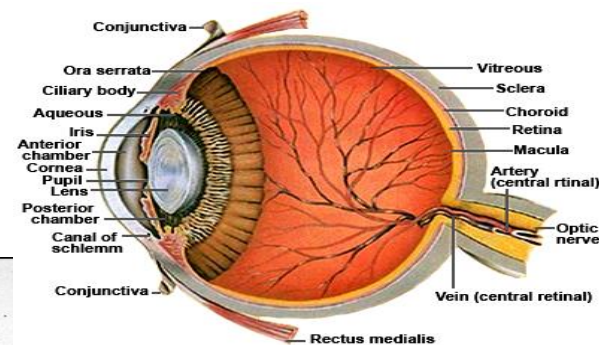
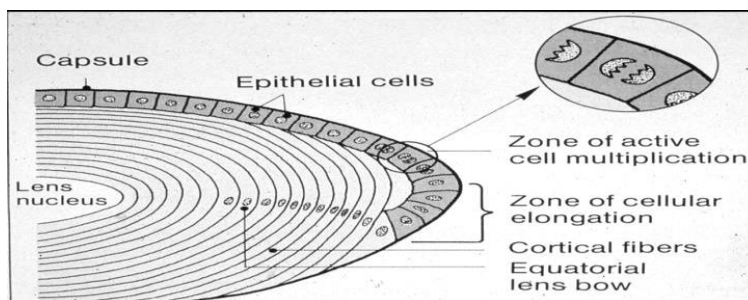
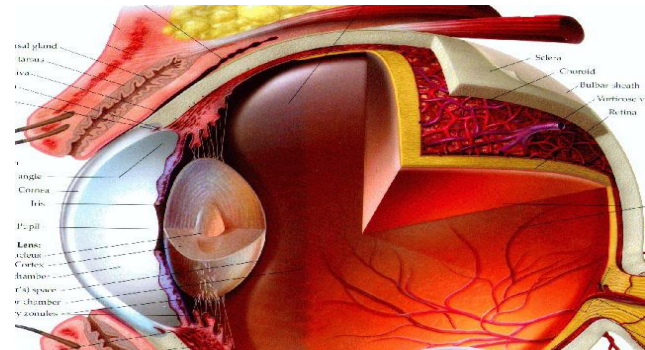
- 1-It gives the color of the eye.
- 2-The pupillae muscle controls size of the pupil:
 - a) **Sphincter pupillae muscle** forms a circular band around the pupil. Its contraction causes constriction of the pupil.
 - b) **The dilator pupillae muscle** which lies near the posterior border of the iris. Its contraction causes dilatation of the pupil.
- 3-Heavy pigmentation of the epithelium prevents passage of light into the interior of the eye except through the pupil.



Eye lens:

The lens is a transparent biconvex structure characterized by great elasticity. It is attached to the ciliary body by a ligament called zonule. It has 3 components:

- 1-Lens capsule: which is rich in collagen and glycoprotein.
- 2-Subcapsular epithelium formed of single layer of cubical cells on the anterior surface of the lens.
- 3-The lens fibers: originated from the epithelial cells that lose their nuclei and organoids. They are elongated fibers and contain crystalline protein. The zonular fibers form the suspensory ligament of the lens.



Vitreous body:

It is a gel nonvascular mass that contains water, collagen, and hyaluronic acid. It occupies region between the lens and retina. It is traversed by the hyaloid canal.

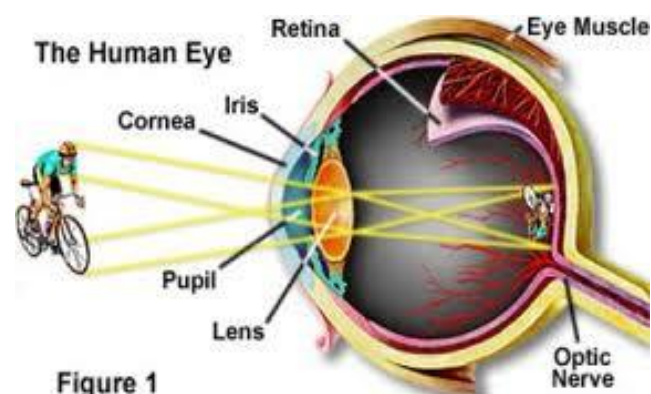
Function:

- 1-Transmission of light to the retina.
- 2-Supports the lens and the retina
- 3-Allows transfer of materials from and to the retina.

Refractive media of the eye:

There are four transparent components through which the light passes till it reaches the retina.

- 1-The cornea. The most effective one.
- 2-The aqueous humor.
- 3-The lens.
- 4-The vitreous body.



III) The inner nervous coat (The retina)

Retina: is the third and innermost layer contains photoreceptor cells.

It is developed from optic cup “evagination of diencephalon, which gives rise to primary optic vesicle”. It is formed of two layers:

A. Outer pigmented epithelium: derived from outer layer of optic cup.

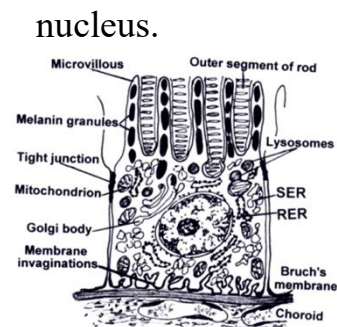
B. Inner nervous layer: derived from inner layer of optic cup.

Histologically retina formed of 10 layers

1. Pigmented epithelium: Columnar cells with basal nucleus. Basal regions adhere firmly to Bruch's membrane. Ion-transporting activity “cell membranes have numerous basal invaginations and mitochondria more abundant near invaginations”.

Lateral membranes show junctional complex. Apical surface has abundant extensions of two types: slender microvilli containing melanin pigment and cylindrical sheaths envelop tips of photoreceptors.

EM: sER, r ER, supra nuclear Golgi, lysosomes, and residual bodies.



Functions

A. Absorption of light and prevention of its reflection.

B. Storage and release of vitamin A “precursor of rhodopsin”.

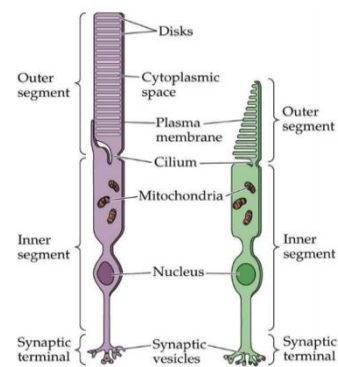
C. Phagocytosis of outer segment of rods during turnover.

2. Rods and cones: Modified dendrites of photoreceptor cells.

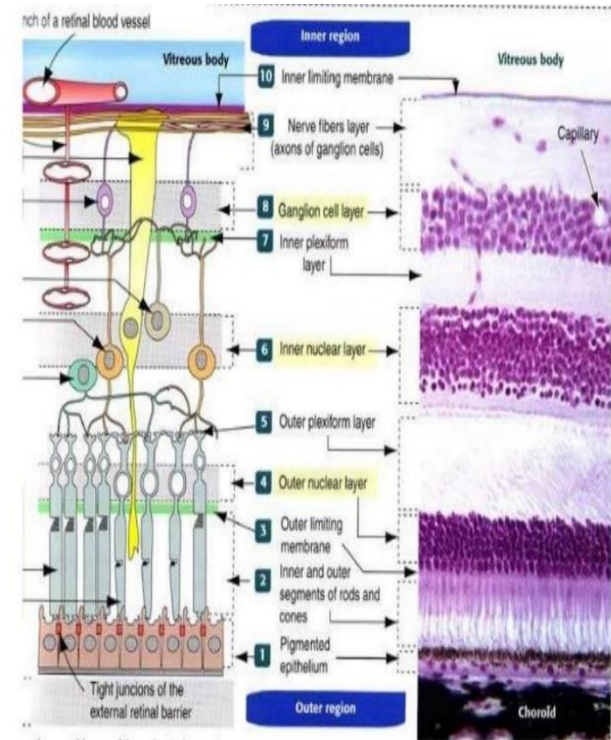
Rods:

Slender bipolar formed of cell body, one dendrite and axon. Dendrite called rod proper and embedded in microvilli of pigment cells “consist of outer and inner segments connected by thin eccentric strand of cytoplasm containing cilium”.

- **Outer segment:** contains transverse membranous discs containing rhodopsin (visual purple), sensitive to dim light “turnover, engulfed by retinal pigment cells and replaced by inner segment”.



- **Inner segment**
Numerous mitochondria, rER ribosomes and Golgi complex.
Responsible for protein synthesis necessary for regeneration of outer segment.
- **Cell body** of rod present in outer nuclear layer.
- **Axon** ends in knob called rod spherule, which synapses with dendrite of bipolar cells.
- **Cones:**
Similar in structure to rods but differ in **Outer segment** conical in shape.
- Tips of disc constant “no turnover”.
- Visual substance “iodopsin” sensitive to bright light.
- Outer cone fiber short.
- Ends in expansion called cone pedicle.



3. Outer limiting membrane

Thin acidophilic layer.

Not true membrane but represents line of cellular junctions between Muller cells and photoreceptors.

4. Outer nuclear layer

Contains cell bodies of rod and cone (First order neuron)

5. Outer plexiform layer

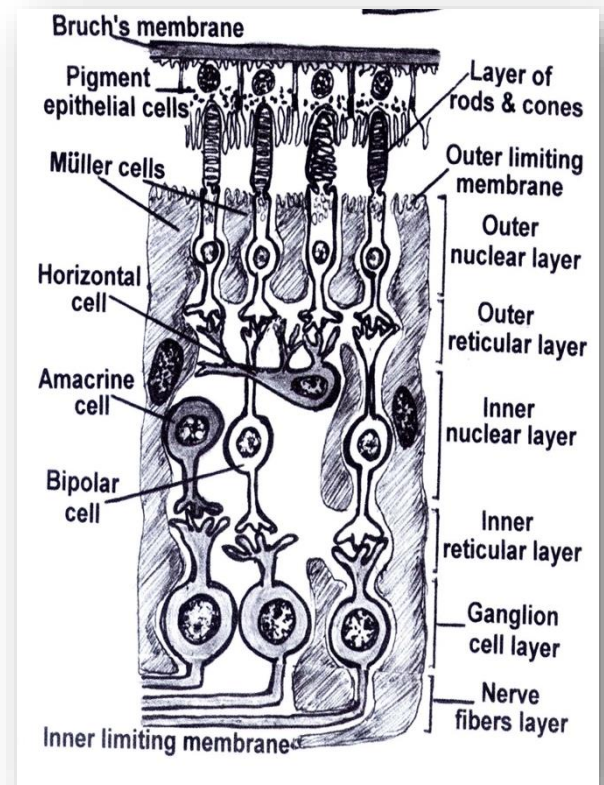
Contains synapse between axons of photoreceptors and dendrites of bipolar cells and horizontal cells.

6. Inner nuclear layer

Formed of nuclei and cell bodies of bipolar cells, amacrine cells, horizontal cells, and

Muller cells (Second order neuron).

- **Bipolar cells:** numerous cells. Dendrites synapse with rods and cones while their axons synapse with ganglionic cells in inner plexiform layer.
- **Horizontal cells:** Situated close to terminal expansion of rods and cones. Their processes run horizontally and extend to outer plexiform layer and synapse with rods and cones.
- **Amacrine cells:** Situated close to dendrites of ganglion cells and synapse with their dendrites.
- **Müller cells:** neuroglial cells extend from outer to inner limiting membranes. Supportive.



7. Inner plexiform layer

Contains processes and synapse between axons of cells present in inner nuclear layer (bipolar and amacrine cells) with dendrites of ganglion cells.

8. Ganglion cell layer

Contains single layer of ganglion cells. Basophilic cytoplasm and vesicular nuclei.

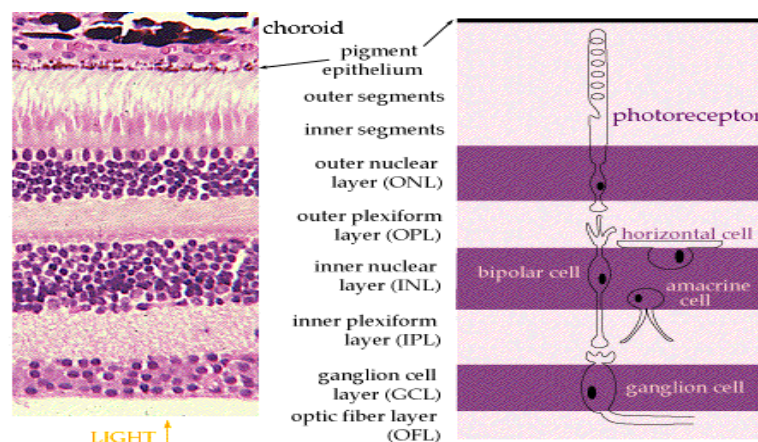
Dendrites synapse with bipolar and amacrine cells in inner plexiform layer and axons form nerve fiber layer.

9. Nerve fiber layer

Unmyelinated axons of ganglion cells pass at right angle to form optic nerve.

10. Inner limiting membrane

Terminal processes of Müller cells and their basal laminae resting on vitreous body. True membrane.



Macula and fovea centralis

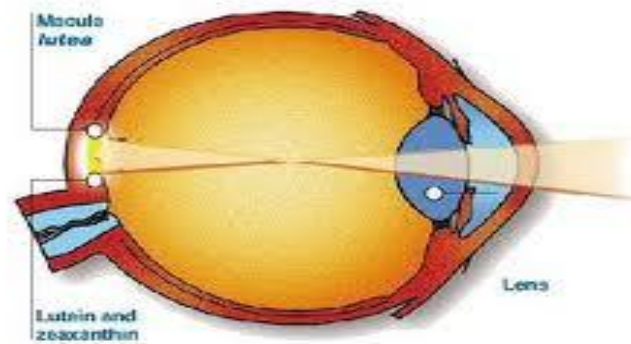
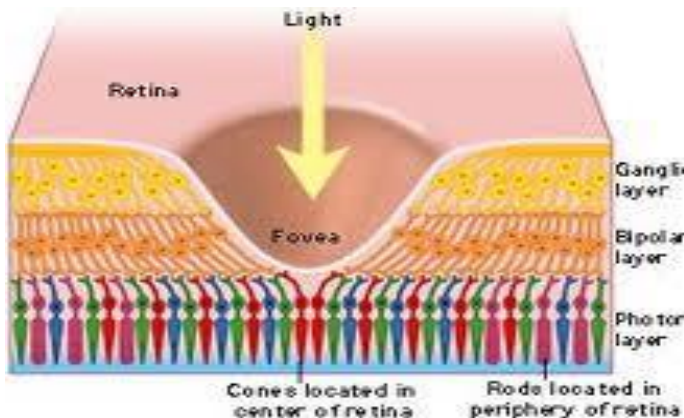
A. Macula lutea

Small yellow area at center of posterior part of retina.

B. Fovea

Thin zone exclusively of cones.

Each cone synapse with only one bipolar cell and linking with single ganglion cell, thus producing high degree of resolution and high detail color vision.



The blood- retinal barriers

- 1) **The outermost barrier:** formed of **tight junction** between the pigmented epithelium of the retina. It prevents the escape of the foreign bodies from the capillaries of the choroid to the outer part of the retina.
- 2) **The innermost barrier:** is formed by the **tight junction** between the endothelial cells of the retinal blood capillaries.

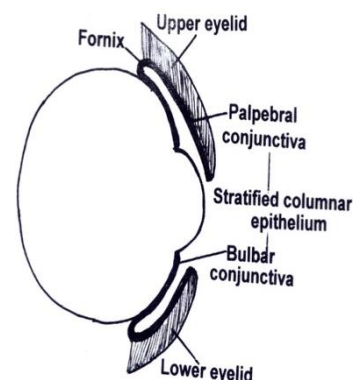
Accessory components of eye

A. Conjunctiva: Thin translucent membrane

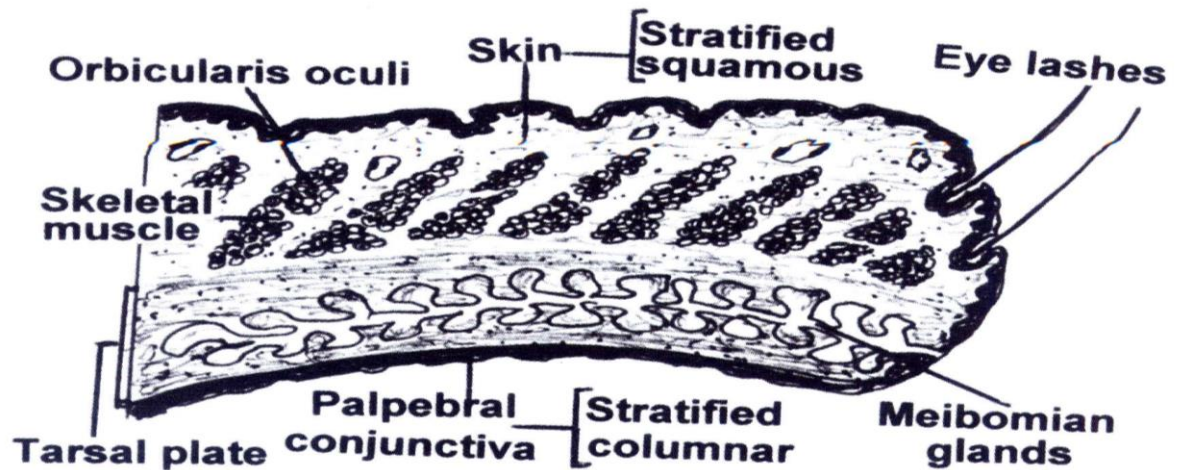
a) **The bulbar conjunctiva:** That covers the anterior part of the sclera up to the cornea. It is formed of **stratified squamous non-keratinized epithelium** with underlying loose connective tissue.

b) **The Palpebral conjunctiva:** It covers the inner surface of the eye lid. It is formed of **stratified columnar epithelium with few goblet cells**.

Function: Lubrication and protection of the eye by the mucous secretion of the goblet cells.



B. Eye lid: Consists of:



1. **Skin containing**
 - a- Eyelashes at margin.
 - b- Glands of Zeis: sebaceous glands open in hair follicle.
 - c- Glands of Moll: sweat glands open between eyelashes.
2. **Orbicularis** muscle: skeletal.
3. **Tarsus:** plate of dense fibro-elastic tissue containing Meibomian glands
“modified sebaceous glands open in lid margin, secrete oily substance delays evaporation of tear film protecting cornea”.
4. **Lining conjunctiva** (palpebral conjunctiva).

C. Lacrimal glands: Tear-secreting gland. Located in anterior superior temporal portion of orbit. Tubulo-alveolar similar in structure to parotid (serous with apical acidophilia and basal basophilia). Excretory ducts connect gland to superior conjunctival fornix.

